DD-Mamba: A Dual-Domain State-Space Forecasting Framework with Linear Anchors and Zero-Initialized Corrections

Anonymous Submission

ADMA 2026 double-blind review

Abstract. Long-term multivariate time series forecasting must reconcile two complementary views: local, trend-driven dynamics in the time domain, and global periodic structure most compact in the frequency domain. Existing state-space (Mamba) forecasters model one view but not both, and deep forecasters often discard simple linear maps that remain near state-of-the-art. We present DD-Mamba, a dual-domain forecasting *framework* whose time branch (decomposition with a selective state-space encoder) and frequency branch (learned spectral filtering) each produce a forecast, fused by a convex gate that preserves both prediction paths. Two principles organize the framework: (i) *linear anchoring* — the time branch carries a DLinear-style linear pathway, the frequency branch an optional FITS-style spectral map, and all nonlinear heads are zero-initialized, so at initialization the model reduces to a linear map of the instance-normalized window; and (ii) *capacity placement* — a bidirectional Mamba over the *variate* dimension carries the model on high-channel data and is disabled on few-channel data where it overfits. On nine benchmarks under the look-back-96 protocol (three seeds), DD-Mamba reports the lowest mean MSE on seven, improving over reported S-Mamba results on five (up to 17% on Solar), tying three within noise, and trailing one, with fewer parameters on small datasets. Paired three-seed ablations on three datasets validate capacity placement in both directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar), show a significant frequency-branch contribution on Solar with redundancy elsewhere, and find zero initialization accuracy-neutral; dual-domain modeling is a per-dataset structural option, not a universal gain.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [13, 8, 14], patch-based attention

[6], inverted variate-token attention [4], and, most recently, selective state-space models (SSMs) built on Mamba [1, 7], which offer linear-time sequence modeling with input-dependent dynamics.

Two empirical facts complicate this progress. First, *simple linear maps remain stubbornly competitive*: a per-component linear projection from the look-back window to the horizon (DLinear [12]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [10]) match or beat far larger models on several benchmarks, especially the ETT family. Deep architectures that ignore this fact spend their early training re-discovering a linear map, and on small datasets often never do. Second, *the useful inductive bias differs by domain*: local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain. Models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [11] — systematically underuse the other.

We address both issues with DD-Mamba, a dual-domain forecaster (Fig. 1) organized around three ideas.

Dual branches with structurally preserved paths. A time branch and a frequency branch each output a complete forecast; the model's prediction is a learned *convex combination* of the two, with the gate conditioned on both branches' features. The output is constrained elementwise between the branch forecasts, so both domains retain an explicit prediction path, and the gate value reports the convex weight assigned to each branch — a proxy for, not a causal measure of, their contributions — while the *actual* contribution of each branch is learned, and the gate may concentrate almost entirely on one branch for some samples or datasets (it is a sigmoid and can saturate near 0 or 1). Our paired ablations find the frequency branch's contribution significant on one of three ablated datasets (Solar) and within noise on the other two (Sect. 4.3); we accordingly claim per-dataset flexibility, not a universal dual-domain gain.

Linear anchors with zero-initialized corrections. Each branch carries an explicit *linear* pathway that parameterizes a strong linear forecaster family of its domain: DLinear-style seasonal/trend window-to-horizon projections [12] in the time branch, and an optional FITS-style complex-linear spectral map [10] in the frequency branch. These are parameterized pathways trained jointly with the model — not pretrained or fitted linear predictors. All nonlinear heads (and the spectral map) are zero-initialized and the fusion gate opens as a constant g_0, so at initialization the model reduces to g_0 times a randomly initialized DLinear-style map of the *instance-normalized* window — linear in the normalized space, the same normalization wrapper used by RLinear-class linear models [3, 2] (Sect. 3.2). We adopt this for the interpretable linear start; a controlled comparison against standard random initialization finds it accuracy-neutral on the two datasets tested (Sect. 4.3), so we position it as an interpretability default, not an accuracy mechanism.

Mamba where it pays. A causal Mamba encoder models the time axis; a *bidirectional* Mamba (frequency bins and channels have no arrow of time) serves two optional roles: spectral encoding, and cross-channel mixing over the variate dimension in the spirit of S-Mamba [7]. Crucially, capacity is *placed* per dataset, and the paired ablations support this in both directions: removing the variate mixer hurts weather, while *adding* it hurts ETTh1 — each with a confidence interval excluding zero (Sect. 4.3).

Our contributions are:

1. A dual-domain forecasting architecture whose output is a gated convex combination of per-branch forecasts, structurally preserving a prediction path from each domain and exposing each branch's per-sample convex weight (Sect. 3).
2. A linear-anchoring scheme — domain-appropriate linear pathways with zero-initialized nonlinear corrections — under which the model at initialization reduces to a linear map of the instance-normalized window; a controlled comparison against random initialization finds it accuracy-neutral, so its value is the interpretable linear start (Sect. 3.2, Sect. 4.3).
3. A capacity-placement recipe for selective SSMs: causal Mamba over time where temporal microstructure matters, bidirectional Mamba over variates where cross-channel structure dominates, and neither where data is scarce (Sect. 3.4).
4. The lowest reported average MSE on seven of nine standard benchmarks under the unified look-back-96 protocol (three-seed mean $\pm$ std for every entry), relative to results published under the same protocol, plus a released ablation harness and paired multi-seed ablations on three datasets — including negative and dataset-dependent findings (instance normalization helps weather and traffic yet hurts solar) that we believe are as actionable as the positive ones (Sect. 4).

2 Related Work

Transformer forecasters. Informer [13], Autoformer [8] and FEDformer [14] adapt attention to long sequences via sparsity, decomposition, and frequency-domain mixing respectively. PatchTST [6] treats sub-series patches as tokens with channel independence; iTransformer [4] inverts the axes and attends over *variates*, establishing that cross-channel modeling, not temporal attention, drives accuracy on high-dimensional benchmarks. DD-Mamba adopts the variate-mixing insight but implements it with linear-time bidirectional SSMs and, unlike these works, couples it to an explicit frequency branch and linear anchors.

Linear and frequency-domain models. DLinear/NLinear [12] showed that decomposition plus a linear map rivals Transformers, particularly on ETT; RLinear [3] added reversible instance normalization [2]. FreTS [11] learns MLPs in the frequency domain, and FITS [10] forecasts with a single complex linear interpolation of the low-passed spectrum at negligible parameter cost. DD-Mamba does not compete with these models — it *contains* them: each branch carries a linear pathway of its domain's model family, and the deep machinery is trained as a correction on top.

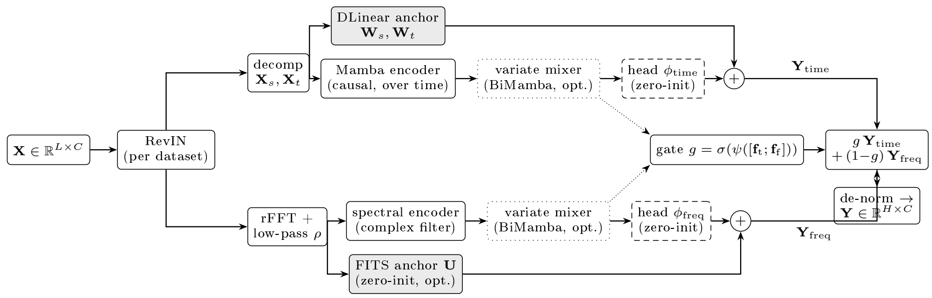

Fig. 1. DD-Mamba overview. Both branches carry a linear anchor (shaded) and zero-initialized nonlinear heads (dashed), so at initialization the model reduces to a linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted arrows carry branch features to the fusion gate, whose convex weight g mixes the two branch forecasts.

State-space models. Mamba [1] introduced selective SSMs with input-dependent dynamics and linear-time scans. In forecasting, S-Mamba [7] applies bidirectional Mamba over variate tokens with a linear temporal embedding, and further works explore frequency-domain SSMs [5]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and over variates, with the placement of each governed by dataset characteristics rather than applied uniformly.

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [2], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [8, 12] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \tag{1}$$

i.e., a DLinear-style map [12] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [1]. A Mamba layer discretizes a state-space system with input-dependent parameters $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$:

$$\mathbf{h}_k = e^{\boldsymbol{\Delta}_k \mathbf{A}}\, \mathbf{h}_{k-1} + \boldsymbol{\Delta}_k \mathbf{B}_k\, u_k, \qquad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D\, u_k, \tag{2}$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\boldsymbol{\Delta}_k \in \mathbb{R},\ \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps [1]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e \tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [7]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{3}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express. We compute the recurrence of Eq. (2) in fp32 even under mixed-precision training: the exponential and the L-step state accumulation overflow in half precision, which silently destroys training (observed as NaN validation losses that exhaust early-stopping patience).

3.3 Frequency Branch: Spectral Filtering and a FITS Anchor

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C \times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top \in \mathbb{C}^{C \times F}$ with $\mathbf{W} \in \mathbb{C}^{F \times F}$ acting along the frequency axis, realized with two real matrices; a bidirectional Mamba over the bin sequence is available as an input-dependent alternative. The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C \times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS anchor (optional, per dataset). Mirroring the time branch, a complex linear map $\mathbf{U} \in \mathbb{C}^{F' \times F}$ with $F' = \lfloor (L{+}H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C \times F'}$), maps the (low-passed) window spectrum to that of a length-$(L{+}H)$ series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \qquad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \tag{4}$$

This pathway *parameterizes* the FITS [10] family — any linear-in-frequency forecaster of the low-passed spectrum is reachable — but, unlike FITS itself, $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization rather than a functioning spectral predictor. This is deliberate: the frequency grids of a length-L and a length-$(L{+}H)$ window do not align (harmonic k of the input maps to the non-integer index $k(L{+}H)/L$ of the output), so no canonical identity or interpolation initialization exists; a randomly initialized $\mathbf{U}$ would inject noise into the fused forecast during the highest-learning-rate epochs. Zero initialization instead lets the gate lean on the time anchor until $\mathbf{U}$ is learned jointly with the rest of the model. Consequently, the precise initialization claim is: the time branch coincides with a (randomly initialized) DLinear, the frequency branch outputs zero, and the fusion gate is the constant g_0 (Sect. 3.5), so in the instance-normalized space the end-to-end map is the linear function $g_0 \cdot \mathbf{Y}_{\text{lin}}$. Composed with RevIN — whose window statistics are themselves a nonlinear function of the input — the model at initialization is a randomly initialized linear map wrapped in instance normalization, exactly the structure of RLinear-class models [3, 2]; it is linear in the normalized window, not in the raw input, and it is linear by construction, not a fitted linear model. The anchor is enabled per dataset (ETTh1, ETTh2, and Weather; Table 2); where it is disabled, the frequency branch remains fully active through its spectral encoder and head but carries no explicit linear pathway, so the linear-anchoring property then rests on the time branch alone.

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [6, 12] but leaves accuracy on the table when variates are strongly coupled [4, 7]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the variate dimension — treating the C per-channel features as a sequence — followed by a position-wise feed-forward block. Bidirectionality is essential: channel order carries no causal direction, so each layer runs forward and reversed scans and sums them in one residual stream. The mixer is a per-dataset choice: $M{=}2$ at width 512 on electricity/traffic (where it is the single most load-bearing component; Sect. 4.3), $M{=}2$ at width 256 on weather, $M{=}1$ on solar, and $M{=}0$ on the 7–8-channel ETT and exchange datasets, where it triples parameters without adding signal.

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])\big) \in (0,1)^{C\times 1}, \quad (5)$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch's convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little for some samples or datasets — which is by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with bias 2.2, so training starts at $g \approx 0.9$: the time branch's output is linear (in the normalized window) at initialization while the frequency branch is silent, and an even 0.5/0.5 mix would waste the highest-learning-rate epochs compensating for noise.

Table 1. Benchmark datasets and where DD-Mamba places its capacity.

Dataset	Variates	Steps	Granularity	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	off	0.24–0.36M
Weather	21	52,696	10 min	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	M=2, d=512	19.7M
Exchange	8	7,588	1 day	off	0.09M

3.6 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$), which concentrates learning in the first epochs and suppresses the late-epoch overfitting that cosine schedules invite on small benchmarks. All runs use 10 epochs; early stopping monitors validation loss with the best checkpoint restored.

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [4, 7]: look-back L=96 for all datasets and horizons $H \in \{96, 192, 336, 720\}$; ETT uses the canonical 12/4/4-month border split (the file tail beyond month 20 is unused), the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE on the standardized test set.

Baselines. We compare against the strongest published results under the same protocol: the state-space forecaster S-Mamba [7], iTransformer [4], PatchTST [6], DLinear [12], and TimesNet [9]. Baseline numbers are quoted from [7] rather than reproduced: the evaluation protocol (look-back, splits, standardization, metrics) is identical, but training pipelines, seeds, and implementations differ across works, so sub-noise-floor gaps to quoted numbers should be read with caution. We flag every such case in the text, and re-running all baselines within one unified training framework is planned for the camera-ready appendix.

Implementation and Reproducibility. All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090 (32 GB). Optimization is AdamW with gradient clipping at 5.0, 10 epochs with the learning rate halved after every epoch, early stopping on validation loss with the best checkpoint restored, and mixed precision with the selective scan forced to fp32 (Sect. 3.2). The decomposition kernel is 25; Mamba blocks use expansion 2 and depthwise convolution width 4 throughout. Table 2 lists all per-dataset settings; complete configuration

Table 2. Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Sect. 3.3); FITS: the *optional* spectral linear anchor of Eq. (4) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	FITS	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

files and the ablation harness ship with the code. All main results (Tables 3 and 4) are means over *three seeds* (2024–2026); Table 4 reports the per-entry standard deviations, whose maximum across all 36 dataset–horizon entries is 0.005 MSE (Solar, H=720) with a typical value of 0.001–0.003.

Statistical methodology. We use two levels of comparison. *Within our framework* (ablations), every variant shares its seeds with the reference, so we report *paired* per-seed differences with t-based 95% confidence intervals (n=3) and treat a component as supported only when its CI excludes zero. *Against quoted baselines*, pairing is impossible, and we use a conservative unpaired tie threshold of 0.005 MSE. Our in-framework DLinear re-run quantifies why this caution is necessary: re-trained through the identical data pipeline, splits, schedule, and evaluation script on ETTh1 (three seeds), DLinear averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — an implementation gap of +0.006, the same order as several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions, and our headline comparison should be read with that caveat; re-running S-Mamba, iTransformer, and PatchTST in this framework is the planned resolution. DD-Mamba's margin over the in-framework DLinear on ETTh1 (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison, roughly six times the largest observed std.

4.2 Main Results

Table 3 summarizes averages over the four horizons; Table 4 reports DD-Mamba per horizon with per-entry standard deviations. Applying the noise-floor tie rule consistently, DD-Mamba reports the lowest average MSE on seven of the nine datasets. The margin over the best baseline is clear on Solar (0.199 vs. iTransformer's 0.233; −17% vs. S-Mamba, roughly seven times the largest observed std

Table 3. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back L=96 (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [7]. **Bold** marks the best result *and* every result within the run-to-run noise floor (< 0.005, Sect. 4.1) of it, treated as tied best; underline marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	**0.439**/**0.434**	0.455/0.450	0.454/0.447	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	**0.374**/**0.401**	0.381/**0.405**	0.383/0.407	0.387/0.407	0.559/0.515	0.414/0.427
ETTm1	**0.386**/**0.398**	0.398/0.405	0.407/0.410	**0.387**/**0.400**	0.403/0.407	0.400/0.406
ETTm2	**0.280**/**0.325**	0.288/0.332	0.288/0.332	**0.281**/**0.326**	0.350/0.401	0.291/0.333
Weather	**0.248**/**0.276**	**0.251**/**0.276**	0.258/**0.278**	0.259/0.281	0.265/0.317	0.259/0.287
Electricity	**0.168**/**0.263**	**0.170**/**0.265**	0.178/0.270	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	0.438/0.281	**0.414**/**0.276**	0.428/0.282	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/**0.405**	0.367/0.408	0.360/**0.403**	0.367/**0.404**	**0.354**/0.414	0.416/0.443
Solar	**0.199**/**0.260**	0.240/0.273	0.233/**0.262**	0.270/0.307	0.330/0.401	0.301/0.319

Table 4. DD-Mamba per-horizon results (mean ± std over three seeds), look-back L=96.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.380±.001	0.431±.002	0.474±.004	0.470±.003	0.439
	MAE	0.396±.001	0.424±.001	0.447±.003	0.468±.001	0.434
ETTh2	MSE	0.293±.001	0.368±.003	0.414±.001	0.421±.001	0.374
	MAE	0.344±.002	0.391±.000	0.426±.000	0.441±.001	0.401
ETTm1	MSE	0.320±.001	0.366±.002	0.398±.001	0.462±.001	0.386
	MAE	0.359±.001	0.384±.001	0.406±.001	0.443±.001	0.398
ETTm2	MSE	0.177±.000	0.242±.002	0.302±.000	0.401±.003	0.280
	MAE	0.260±.001	0.303±.001	0.340±.000	0.398±.002	0.325
Weather	MSE	0.162±.000	0.212±.000	0.269±.001	0.352±.003	0.248
	MAE	0.207±.001	0.255±.000	0.295±.002	0.349±.002	0.276
Electricity	MSE	0.140±.000	0.157±.001	0.174±.001	0.200±.004	0.168
	MAE	0.235±.000	0.252±.001	0.271±.001	0.296±.004	0.263
Traffic	MSE	0.402±.002	0.424±.001	0.441±.001	0.485±.003	0.438
	MAE	0.266±.001	0.275±.001	0.283±.000	0.301±.000	0.281
Exchange	MSE	0.086±.000	0.179±.001	0.330±.002	0.857±.001	0.363
	MAE	0.205±.000	0.301±.000	0.416±.002	0.699±.000	0.405
Solar	MSE	0.180±.004	0.195±.005	0.207±.001	0.212±.005	0.199
	MAE	0.237±.003	0.259±.003	0.270±.001	0.274±.005	0.260

— measured with the instance-normalization-free configuration from Sect. 4.3), on ETTh1 (−0.015), and on ETTh2 (−0.007); on ETTm1/ETTm2 the margin over S-Mamba is clear (−0.012/−0.008) but the gap to PatchTST is 0.001 — a tie; on Weather (−0.003) and Electricity (−0.002) the lead over S-Mamba is below the floor and the methods should be read as tied. DD-Mamba remains competitive on Exchange — a near random-walk dataset where all methods cluster within 0.013 MSE and plain DLinear is best, 0.009 ahead of DD-Mamba — and lags S-Mamba on Traffic by +0.024 MSE. We *hypothesize* the traffic deficit stems from the weekly cycle exceeding the protocol look-back; this is not yet experimentally verified, and Sect. 5 lists the experiments that would test it. All comparisons are relative to results *reported* in [7] under the same protocol (Sect. 4.1), so we phrase our overall claim conservatively: DD-Mamba is competitive with results reported under the same protocol. Its largest margin is on Solar, driven by per-dataset normalization; the next largest are on the

Table 5. Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; * CI excludes zero. The FITS anchor is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802$\pm$.001	0.1802$\pm$.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
FITS anchor removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

small ETT datasets, where the linear anchors and deliberate capacity restraint matter most — there DD-Mamba uses roughly 0.25M parameters, two orders of magnitude fewer than variate-attention models configured for those datasets.

4.3 Component Ablations

We release an ablation harness in which every variant flips exactly one switch of a dataset's full configuration, holding the split, schedule, seeds, and all other hyperparameters fixed. We report its complete output on three datasets under their *final released configurations*, three seeds each and all at H=96: ETTh1 and Solar-Energy (Table 5) and Weather (Table 6), analyzed with the paired methodology of Sect. 4.1. Two internal consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry exactly (0.180$\pm$.004), and Solar's w/o-instance-norm variant coincides with its full configuration (which already disables RevIN), so that cell is omitted.

Dual-domain evidence. The decisive result is on Solar: removing the frequency branch degrades MSE by +0.0114 with a CI excluding zero — the first statistically supported frequency-branch contribution in our studies. On ETTh1 and Weather, by contrast, both single-branch variants lie within their CIs; on ETTh1 the frequency branch *alone* even matches the full model (−0.0007), i.e., the two branches are largely redundant there. The dual-domain claim our data supports is therefore conditional: the frequency path contributes measurably on Solar and is redundant, though not harmful, on the other two ablated datasets.

Capacity placement, validated in both directions. Adding the variate mixer on ETTh1 *hurts* (+0.0077, CI [+0.004, +0.012]), while removing it on Weather also hurts (+0.0104, [+0.004, +0.017]) — significant paired evidence on both sides of the placement decision. Instance normalization shows the same two-sided pattern: removing it hurts Weather (+0.0161, [+0.005, +0.027]), while *disabling*

Table 6. Component ablation on Weather (H=96; MSE mean ± std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620±.0003	—	reference
w/o instance norm	0.1781±.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724±.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647±.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631±.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628±.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o FITS anchor	0.1627±.0016	+0.0007 [−0.0041, +0.0055]	spectral linear anchor
freq branch only	0.1626±.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626±.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622±.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

it on Solar improved the re-measured main results from 0.228 to 0.199 average MSE (a legacy single-seed probe on the pre-final Solar configuration first surfaced this as −0.016 MSE at H=96).

Zero initialization: a null result. The controlled random-initialization comparison — the harness switch introduced precisely for this test — finds no significant accuracy difference on either dataset (−0.0012 on ETTh1, −0.0046 on Solar; both CIs include zero). We therefore make no accuracy claim for zero initialization: its value in this framework is the interpretable linear start of Sect. 3.2, and stability-oriented analyses (convergence curves, failure rates, correction magnitudes) remain future work.

Weaker components. The DLinear backbone is not individually significant in any paired study (point estimates +0.008 on Solar, +0.001 on Weather, −0.003 on ETTh1, where the FITS pathway can substitute for it), and the learned gate versus plain averaging is within CIs everywhere. On Weather (Table 6), exactly three components survive the paired test: instance normalization (+0.0161), the variate mixer (+0.0104), and the temporal Mamba (+0.0027, [+0.0020, +0.0034], positive on all three seeds); every other variant's CI includes zero. Across the three studies the recurring pattern is that component value is dataset-dependent — in *both* directions for the mixer and normalization — which is the argument for shipping components as per-dataset switches rather than universal defaults.

Normalization is a per-dataset decision. The same probe that revealed RevIN *hurting* solar (−0.016 MSE when removed) shows the opposite on traffic: removing instance normalization degrades traffic from 0.405 to 0.505 MSE at H=96. The two datasets look alike on the surface — both bounded, both strongly periodic — but solar's night zeros are *periodic structure* (window statistics then mostly encode the window's night fraction, and de-normalization amplifies errors by an unstable standard deviation), whereas traffic windows carry genuine distribution shift (weekday/weekend composition, per-sensor levels) that instance normalization absorbs. We therefore ship normalization as a verified per-dataset switch and caution against transferring normalization conclusions across datasets.

Negative results that shaped the design. Deepening the variate mixer on traffic from two to four layers (19.7M $\rightarrow$ 37.9M parameters) produced no accuracy gain, so the released configuration keeps two layers; our working *hypothesis* for the residual gap to S-Mamba on traffic is the weekly cycle, which at hourly sampling (168 steps) exceeds the protocol look-back of 96 and is invisible to both branches — a hypothesis directly testable by widening the look-back to 168/336, adding calendar covariates, or re-comparing against S-Mamba under a longer window, none of which we have run yet. Similarly, enabling the variate mixer on ETT (7 channels, 8.4K training windows) tripled parameters and *caused* premature early stopping through immediate overfitting; the released ETT configurations disable it.

4.4 Analysis

Where each domain wins. The branch-only ablations (Table 5) quantify the two domains directly on solar: removing the time branch costs +0.012 MSE while removing the frequency branch costs +0.002, i.e., both contribute but the time view carries more — consistent with the protocol look-back hiding part of the periodic structure (at 10-minute resolution the daily cycle spans 144 samples $> L$=96; the same argument applies to weather's daily and traffic's weekly cycles). Beyond ablations, the fusion gate (Eq. (5)) exposes per-sample, per-channel convex mixing weights at no extra cost — a proxy for, not a causal measure of, branch contributions; a systematic study of gate behavior across datasets is left for the extended version.

Efficiency. We make no measured efficiency claim in this version and note only two design properties: parameter counts span 0.09M–19.7M across datasets (Table 1), sized to each dataset rather than fixed; and the bidirectional variate mixer scans C tokens rather than $B \times C$ folded time sequences, so its cost grows with the channel count, not with batch $\times$ channels, while the causal time scan is enabled only where the channel count keeps it affordable. A wall-clock, memory, and throughput comparison against S-Mamba, iTransformer, and PatchTST under identical hardware is left for the extended version.

5 Conclusion

DD-Mamba is a dual-domain state-space forecasting *framework*: each branch carries a linear pathway of its domain, deep components are zero-initialized corrections, and capacity (especially cross-variate mixing) is placed per dataset rather than uniformly. Under the unified look-back-96 protocol, this recipe achieves the lowest reported average MSE on seven of nine standard benchmarks and remains competitive on Exchange, while still lagging behind S-Mamba on Traffic, with parameter counts spanning 0.09M to 19.7M across datasets. We state plainly what the paired ablations do and do not support: capacity placement is validated in *both* directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar), the temporal encoder helps weather, and the frequency branch contributes measurably on Solar while being redundant on ETTh1 and weather; zero initialization is accuracy-neutral in a controlled comparison. We therefore claim per-dataset structural flexibility with dataset-dependent branch value, not a universal benefit of combining domains. They also surface a transferable methodological caution: instance normalization can help or hurt substantially depending on whether apparent nonstationarity is real shift or periodic structure.

Limitations remain; we list each with the experiment that would resolve it. (1) The frequency-branch gain is demonstrated on a single dataset (Solar) at one horizon with n=3; extending the paired matrix across horizons and the remaining datasets would establish where it generalizes. (2) Zero initialization is accuracy-neutral in our controlled comparison, but its hypothesized stability value remains untested — convergence curves, failure rates, and the magnitude of the learned corrections are open analyses. (3) Baseline numbers are quoted from [7] rather than re-run; the in-framework DLinear re-run (Sect. 4.1) exposes a +0.006 implementation gap, so comparative claims should be read as "competitive with reported results under the same protocol" until S-Mamba, iTransformer, and PatchTST are reproduced in this framework. Architecturally: the fixed look-back of 96 hides cycles longer than the window (weather's daily and traffic's weekly periods); we hypothesize this explains the remaining deficit on traffic, a claim to be tested with longer look-backs (168/336), calendar covariates, and a same-window re-comparison against S-Mamba. Multi-resolution windows and adaptive spectral cutoffs are natural next steps.

Reproducibility. Code, configurations, and the ablation harness are provided as supplementary material and will be released in a public repository upon acceptance.

References

1. Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv:2312.00752 (2023)
2. Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.H., Choo, J.: Reversible instance normalization for accurate time-series forecasting against distribution shift. In: ICLR (2022)
3. Li, Z., Qi, S., Li, Y., Xu, Z.: Revisiting long-term time series forecasting: An investigation on linear mapping. arXiv:2305.10721 (2023)
4. Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., Long, M.: iTransformer: Inverted transformers are effective for time series forecasting. In: ICLR (2024)
5. Ma, S., Kang, Y., Bai, P., Zhao, Y.-B.: FMamba: Mamba based on fast-attention for multivariate time-series forecasting. arXiv:2407.14814 (2024)
6. Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J.: A time series is worth 64 words: Long-term forecasting with transformers. In: ICLR (2023)

7. Wang, Z., Kong, F., Feng, S., Wang, M., Zhao, H., Wang, D., Zhang, Y.: Is Mamba effective for time series forecasting? Neurocomputing (2025)
8. Wu, H., Xu, J., Wang, J., Long, M.: Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In: NeurIPS. pp. 22419–22430 (2021)
9. Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., Long, M.: TimesNet: Temporal 2D-variation modeling for general time series analysis. In: ICLR (2023)
10. Xu, Z., Zeng, A., Xu, Q.: FITS: Modeling time series with 10k parameters. In: ICLR (2024)
11. Yi, K., Zhang, Q., Fan, W., Wang, S., Wang, P., He, H., An, N., Lian, D., Cao, L., Niu, Z.: Frequency-domain MLPs are more effective learners in time series forecasting. In: NeurIPS (2023)
12. Zeng, A., Chen, M., Zhang, L., Xu, Q.: Are transformers effective for time series forecasting? In: AAAI. pp. 11121–11128 (2023)
13. Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., Zhang, W.: Informer: Beyond efficient transformer for long sequence time-series forecasting. In: AAAI. pp. 11106–11115 (2021)
14. Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., Jin, R.: FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: ICML. pp. 27268–27286 (2022)